Atomic, Quantum and Solid State Physics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Atomic, Quantum and Solid State Physics (A06812)

Short Title: Atomic, Quantum & Solid State
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author GGALLAGHER
Credits 5 **Level:** Intermediate

Description of Module / Aims

In this module the student meets formal quantum mechanics for the first time. Schrodinger's equation is introduced and applied to a range of situations, from the hydrogen atom to the crystal lattice, and the solutions are interrogated to provide information on a range of observables. The module concludes with an introduction to the quantum theory of magnetism.

Programmes

ELEC-0039	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	3 / 5 / M

Indicative Content

- Introduction to wave mechanics: the wave function and Schrodinger's equation; separation of variables; time-independent equation
- Application of Schrodinger's equation to infinite and finite square well potentials
- Scattering and tunnelling
- Probability densities, expectation values and uncertainty; Heisenberg's uncertainty principle
- The hydrogen atom, the quantum numbers, electron wave functions and probability densities; the periodic table
- Electrons in metals: the QM free electron model, the Fermi-Dirac distribution function and the density of states function
- The band theory of solids: the Kronig-Penney model, the tight-binding approximation
- Band structure of metals, insulators and semiconductors
- Diamagnetism, ferromagnetism, paramagnetism and electron paramagnetic resonance
- Simulations: Schrodinger's equation, eigenvalues and eigenfunctions in the lattice

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply Schrodinger's equation to a series of well-defined potential fields generating solutions for the wave function.
2. Use his/her knowledge of the wave function to ascertain important properties of the system such as energy, probability of location, expectation values.
3. Apply Schrodinger's equation to the hydrogen atom obtaining solutions incorporating the appropriate quantum numbers.
4. Apply quantum mechanics to electrons in atoms, metals and the crystal lattice.
5. Use simulation methodologies in the investigation and analysis of atomic and solid state phenomena.

Learning and Teaching Methods

- Lectures.
- Tutorials.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
In-Class Assessment	50%	1,2,3
In-Class Assessment	40%	4
Assignment	10%	5

Assessment Criteria

<40%: Inability to understand the basic concepts encountered in this module.
40%-49%: Ability to grasp the fundamental concepts of atomic, quantum and solid state physics.
50%-59%: Ability to understand and explain the fundamental concepts of atomic, quantum and solid state physics and demonstrate some problem-solving skills.
60%-69%: All the above, and in addition, apply problem-solving techniques to advanced problems.
70%-100%: All previous to an excellent level. Demonstrates an ability to put a solution into the context of the course material and assess whether such solutions are meaningful.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	12	
Independent Learning	87	

Essential Material

- Garcia, A. and S. Damask. _Physics for Computer Scientists_. New York: Springer, 1998.

Supplementary Material

- Beiser, A. _Concepts of Modern Physics_. 6th ed. USA: McGraw-Hill, 2003.
- Bolton, J. _Wave Mechanics_. UK: Open University, 2009.
- Griffiths, D.J. _Introduction to Quantum Mechanics_. 2nd ed. UK: Pearson, 2014.